

Finite Arithmetic Capacity under Additive Locally Constant, Good-Reduction Multiplier, and Algebraic-Action Readouts

Anonymous

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Abstract

We study a fixed additive periodic-orbit readout whose exact prime hits admit representations

$$\log p = v_p + \log q_p + \alpha_p,$$

where the v_p lie in one finite-dimensional rational space V , each $q_p > 0$ is algebraic and q_p^2 is a unit away from one finite rational-prime set $S_{\mathbb{Q}}$, and each α_p is real algebraic. We prove the rank-plus-support capacity bound $|\mathcal{P}_{\text{hit}}| \leq \dim_{\mathbb{Q}} V + |S_{\mathbb{Q}}|$. A rational dependence among outside-support v_p terms produces $\log R = \beta$ with $R > 0$ algebraic and β algebraic. Hermite–Lindemann forces $R = 1$; after squaring, a finite-place valuation above each distinct outside prime forces its relation coefficient to vanish. The proof requires neither a prior finiteness assumption nor a target search.

We give source certificates for fixed finite-memory locally constant readouts, good-reduction generalized Hénon multiplier moduli, and regularly evaluated algebraic exact actions. A selector architecture is a corollary of the additive theorem. A source-locked exact/static audit closes nine proof, scope, control, isolation, review, and upstream gates; its single registered run has 51 passing tests, zero numerical candidate runs, and zero target matches. The result is an exact capacity certificate for the declared class, not a universal symplectic no-go theorem, a complete escape trichotomy, or a Riemann-zero construction.

1 Introduction

Prime logarithms arise as desired periodic scales in several arithmetic and spectral analogies. Berry and Keating describe a hypothetical chaotic “Riemann dynamics” whose primitive periods are tied to logarithms of primes (Berry and Keating, 1999). Deninger’s foliated and arithmetic dynamical programs assign length $\log Nx$ to closed-point packets (Deninger, 1998, 2026); the scaling-site and adèle-class-space program of Connes and Consani contains prime-indexed periodic orbits of length $\log p$ (Connes and Consani, 2016, 2024). These works motivate the scale, but they also warn against a universal obstruction: arithmetic spaces can carry the primes intrinsically.

We ask a narrower design question. Suppose a finite architecture combines three familiar sources of periodic observables: a finite-rank locally constant term, logarithms of multiplier moduli with one fixed finite bad-prime support, and an algebraically evaluated action term. How many distinct rational-prime logarithms can the resulting *additive* readout equal exactly when all maps, supports, coefficients, and normalizations are fixed before the target is inspected?

The answer is a rank-plus-support bound. If every hit prime satisfies

$$\log p = v_p + \log q_p + \alpha_p, \quad v_p \in V, \quad q_p^2 \in \mathcal{O}_{S_{\mathbb{Q}}}^{\times}, \quad \alpha_p \in \overline{\mathbb{Q}} \cap \mathbb{R}, \quad (1)$$

with $q_p \in \overline{\mathbb{Q}} \cap \mathbb{R}_{>0}$, then

$$|\mathcal{P}_{\text{hit}}| \leq \dim_{\mathbb{Q}} V + |S_{\mathbb{Q}}|. \quad (2)$$

Here $\mathcal{O}_{S_{\mathbb{Q}}}^{\times}$ is shorthand for zero valuation at all finite places not above $S_{\mathbb{Q}}$; the definition is made field-independent below.

Our contributions are four precise statements.

1. We prove the additive bound (2), including rational and negative multiplier powers, arbitrary valid certificate selection, repeated-hit set semantics, and the $q = 1$ edge case.
2. We supply complete source certificates: higher-block recoding for locally constant readouts; a projective, non-Archimedean, and monodromy argument for good-reduction generalized Hénon maps; and regular algebraic action evaluation with a separate gauge-invariance ledger.
3. We demote the earlier selector union to a corollary and state the escape map only as necessary failures of this certificate. The exits are neither exclusive, exhaustive, nor sufficient for arithmetic provenance.
4. We bind the theorem and scope to a registered exact/static audit. Nine gates pass, 20 proof IDs and 19 scope-operation IDs close, and 51 official tests pass without reading target data or executing a numerical candidate.

The scope restriction is substantive. We do not treat an algebraic irrational coefficient on a multiplier logarithm, an arbitrary nonlinear mixer, a logarithm applied after an algebraic action, a general Hölder roof, a complex logarithm branch, or approximate equality. We make no historical priority claim and no claim about zeta zeros, trace formulas, determinants, quantization, or “Route B.”

2 Arithmetic and dynamical context

Symbolic periods. Shifts of finite type, higher-block presentations, suspension flows, and periodic-orbit zeta functions provide the standard finite-state setting for locally constant roofs (Parry and Pollicott, 1990; Lind and Marcus, 1995). Our Class-L statement uses only an elementary consequence: a fixed finite-memory observable becomes an edge function on a finite higher-block graph. We do not assert finite rational rank for a general point-dependent Hölder roof.

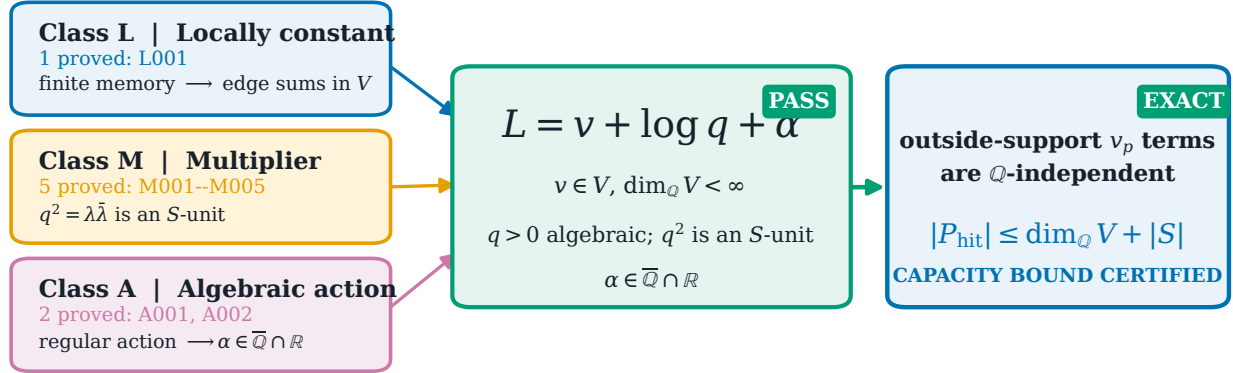
Arithmetic Hénon dynamics and multipliers. Plane polynomial automorphisms decompose into elementary and generalized Hénon pieces (Friedland and Milnor, 1989). Arithmetic studies develop heights, periodic points, and local behavior for Hénon maps (Silverman, 1994; Ingram, 2014; Hsia and Kawaguchi, 2018). Recent multiplier work studies complex rigidity and thermodynamic geometry (Cantat and Dujardin, 2026; Bianchi and He, 2026). Our Class-M certificate does not claim novelty for that broad literature. It isolates the narrower fact that a frozen good-reduction architecture puts squared return moduli in an S -unit group with one fixed rational-prime support.

Exact action readouts. Generating potentials and action sums are standard tools for exact symplectic maps and variational dynamics (Delshams and Ramírez-Ros, 1997; Bialy and Tsodikovich, 2023). In the present certificate, the operative arithmetic fact is regular evaluation: a rational potential over $\overline{\mathbb{Q}}$ evaluated on an algebraic finite orbit gives an algebraic sum. A full endpoint ledger is needed to distinguish preservation of algebraicity from canonical gauge invariance.

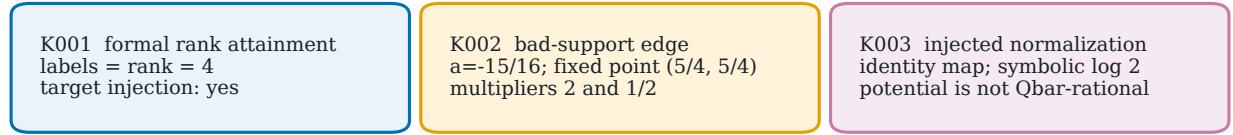
a Certified sources

b Additive form

c Capacity



d Exact controls and scoped escape



certificate failures are necessary only: not exclusive, exhaustive, or sufficient

Figure 1: The scoped additive certificate. A fixed finite-rank term, a fixed-support multiplier-log term, and a real algebraic action term combine in one readout. Hermite–Lindemann and valuation isolation yield the rank-plus-support bound. The lower boundary records necessary ways to leave the certificate; it is not a complete or sufficient trichotomy. The categorical states are loaded from the frozen source and official audit.

Transcendence and arithmetic spaces. The only transcendence input is Hermite–Lindemann, in the form that e^β is transcendental for nonzero algebraic β (Baker, 2022). Standard valuation and extension facts are used in their usual number-field form (Neukirch, 1999). Conversely, Deninger’s infinite-dimensional systems and the Connes–Consani scaling/adèle spaces are positive architectures outside the finite L/M/A class. Connes’s trace-formula program supplies wider arithmetic-spectral context (Connes, 1999), but no trace-formula or zero-spectral conclusion is drawn here.

Table 1: Closest contexts and the claim boundary.

Context	Overlap	Difference here
Symbolic suspensions	Finite-state periodic sums and zeta functions	We prove an exact rational-capacity statement for a fixed finite-rank subspace.
Arithmetic Hénon maps	Good reduction, heights, periodic points	We use a squared-modulus S -unit certificate inside a mixed additive theorem.
Exact actions	Periodic generating sums and gauge choices	We retain only regular algebraic evaluation as one additive source.
Arithmetic spaces	Prime/closed-point orbits of length $\log p$ or $\log Nx$	These positive examples lie outside the finite certificate and preclude universal framing.

3 The additive certificate and its closure

Fix a finite set $S_{\mathbb{Q}}$ of rational primes. For a nonzero algebraic number x , call x an $S_{\mathbb{Q}}$ -unit if, in a number field containing x ,

$$v_w(x) = 0 \quad \text{for every finite place } w \text{ not lying above } S_{\mathbb{Q}}. \quad (3)$$

We saturate the exceptional places above $S_{\mathbb{Q}}$. The definition is unchanged after finite extension.

Lemma 3.1 (Extension invariance). *Let E/K be a finite extension of number fields and $x \in K^{\times}$. If x has zero valuation at every finite place of K away from $S_{\mathbb{Q}}$, then the same holds at every finite place of E away from $S_{\mathbb{Q}}$. Multiplication and inversion preserve the property.*

Proof. If w lies above v , normalized additive valuations satisfy $v_w(x) = e(w/v)v_v(x)$ up to the common positive normalization convention. A place w away from $S_{\mathbb{Q}}$ lies above a place v away from $S_{\mathbb{Q}}$, so both sides vanish. The product and inverse statements follow from additivity. $\square$

Definition 3.2 (Canonical additive readout). Fix a finite-dimensional $\mathbb{Q}$ -subspace $V \subset \mathbb{R}$, the set $S_{\mathbb{Q}}$, and all maps, coefficients, supports, gauges, and assembly operations independently of the eventual target. An in-scope readout is

$$L = v + \log q + \alpha, \quad v \in V, \quad q \in \overline{\mathbb{Q}} \cap \mathbb{R}_{>0}, \quad q^2 \text{ an } S_{\mathbb{Q}}\text{-unit}, \quad \alpha \in \overline{\mathbb{Q}} \cap \mathbb{R}, \quad (4)$$

where $\log$ is the real logarithm on positive inputs.

The target-independence clause is a provenance condition excluding preloaded lookup data. Once valid representations are supplied, the proof below uses only the common spaces V and $S_{\mathbb{Q}}$ and the stated algebraicity and unit conditions; it does not use the mechanism by which provenance was certified.

Lemma 3.3 (Rational multiplier-log closure). *Let $q_1, \dots, q_t > 0$ be algebraic, with every q_j^2 an $S_{\mathbb{Q}}$ -unit, and let $c_j \in \mathbb{Q}$. Then*

$$\sum_{j=1}^t c_j \log q_j = \log q$$

for a positive algebraic q whose square is an $S_{\mathbb{Q}}$ -unit.

Proof. Choose $D \geq 1$ with $m_j = Dc_j \in \mathbb{Z}$ and set $q = \prod_j q_j^{m_j/D}$, using the unique positive real algebraic roots in a finite extension. The real logarithm gives the stated equality. Moreover $(q^2)^D = \prod_j (q_j^2)^{m_j}$. At a place outside $S_{\mathbb{Q}}$, the right side has zero valuation, hence $Dv_w(q^2) = 0$. Negative exponents are harmless because the unit group is closed under inversion. $\square$

Thus the class admits finite rational sums, differences, repetitions, and rational scalings of L terms; rational sums of positive M logarithms; negative and rational M powers; $q = 1$; and finite algebraic sums/scales of real algebraic A terms. It excludes algebraic irrational M-log coefficients, nonlinear post-assembly maps, target/orbit lookup tables, log-after-action, general Hölder roofs without a common finite-rank proof, complex-log branches, approximate matching, and target-dependent support or weights.

4 Finite arithmetic capacity

Let $\mathcal{P}_{\text{hit}}$ denote a set of *distinct* positive rational primes. Repeated orbit hits and repeated certificates do not change its cardinality.

Theorem 4.1 (Additive finite arithmetic capacity). *Suppose every $p \in \mathcal{P}_{\text{hit}}$ has at least one representation*

$$\log p = v_p + \log q_p + \alpha_p \quad (5)$$

within [definition 3.2](#). Select one arbitrary valid representation for each distinct prime before testing any relation. Then

$$|\mathcal{P}_{\text{hit}}| \leq \dim_{\mathbb{Q}} V + |S_{\mathbb{Q}}|. \quad (6)$$

No prior finiteness assumption on $\mathcal{P}_{\text{hit}}$ is required.

Proof. Put $\mathcal{P}_0 = \mathcal{P}_{\text{hit}} \setminus S_{\mathbb{Q}}$. We prove that the selected family $\{v_p : p \in \mathcal{P}_0\}$ is linearly independent over $\mathbb{Q}$.

Take distinct $p_1, \dots, p_k \in \mathcal{P}_0$ and suppose $\sum_i c_i v_{p_i} = 0$ with $c_i \in \mathbb{Q}$. Clearing denominators produces integers m_i , not all zero, such that

$$\sum_{i=1}^k m_i v_{p_i} = 0. \quad (7)$$

Substitution of (5) into (7), using positivity and the real logarithm law even for negative powers, gives

$$\log R = \beta, \quad R = \frac{\prod_i p_i^{m_i}}{\prod_i q_{p_i}^{m_i}} \in \overline{\mathbb{Q}} \cap \mathbb{R}_{>0}, \quad \beta = \sum_i m_i \alpha_{p_i} \in \overline{\mathbb{Q}} \cap \mathbb{R}. \quad (8)$$

If $\beta \neq 0$, Hermite–Lindemann makes e^β transcendental, contrary to $e^\beta = R \in \overline{\mathbb{Q}}$. Hence $\beta = 0$, and injectivity of the real exponential gives $R = 1$.

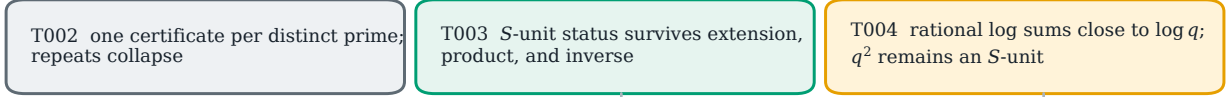
Square this algebraic identity before taking valuations:

$$\prod_{i=1}^k p_i^{2m_i} = \prod_{i=1}^k (q_{p_i}^2)^{m_i}. \quad (9)$$

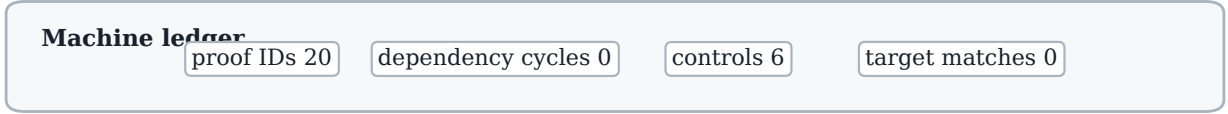
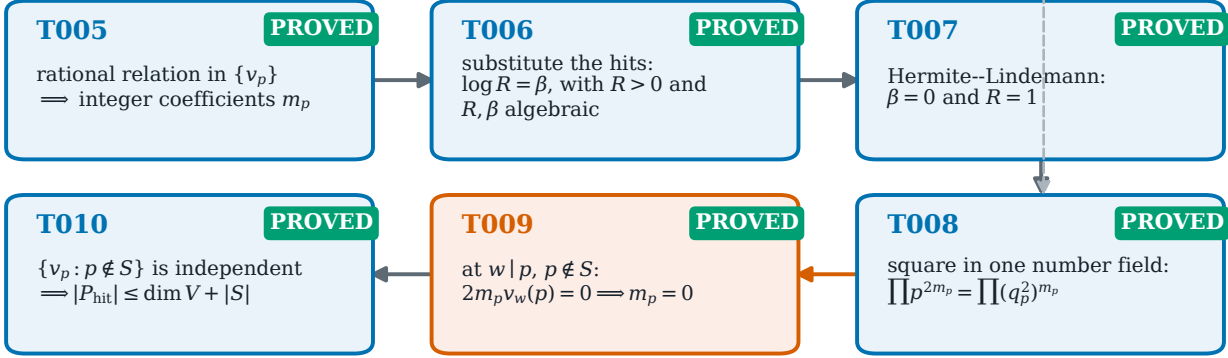
Put the finitely many factors into one number field E . [Lemma 3.1](#) preserves the $S_{\mathbb{Q}}$ -unit condition. Fix i and take a finite place w of E above p_i . The right side of (9) has zero w -valuation because $p_i \notin S_{\mathbb{Q}}$. For $j \neq i$, the distinct rational prime p_j also has zero w -valuation, while $v_w(p_i) > 0$. Therefore $2m_i v_w(p_i) = 0$ and $m_i = 0$. This holds for every i , contradicting a nonzero relation.

Every finite subfamily of the outside v_p terms is independent. A $\mathbb{Q}$ -space of dimension r cannot contain $r + 1$ independent vectors, so $|\mathcal{P}_0| \leq r$. At most $|S_{\mathbb{Q}}|$ distinct hit primes lie inside $S_{\mathbb{Q}}$, which proves (6). If $\mathcal{P}_0$ were infinite, choosing $r + 1$ elements would already give the same contradiction. $\square$

a Locked prerequisites



b Relation-to-capacity proof spine



Highlighted valuation isolation is the arithmetic step that eliminates every nonzero outside-prime coefficient.

Figure 2: The complete proof flow. The only transcendence input removes the real algebraic additive contamination. Squaring then exposes only the certified q_p^2 factors, and one finite place above each distinct outside-support prime isolates its coefficient. The frozen proof ledger records every dependency; the diagram is not a numerical proof.

Remark 4.2 (Edge cases and abstract attainability). If $V = \{0\}$, no outside-support prime can be hit. The case $q_p = 1$ simply adds zero. The rank contribution is abstractly attainable: a graph with r deliberately inserted independent labels realizes r formal labels. This shows capacity of the bare rank hypothesis only. It is target injection, not arithmetic emergence or an intrinsic-provenance optimality result. Likewise, the fixed Hénon control with $a = -15/16$ and fixed point $(5/4, 5/4)$ has exact multiplier moduli 2 and $1/2$ only because the denominator has already put the prime 2 in the bad support.

5 Three source certificates

5.1 Class L: finite memory gives finite rank

Let a fixed finite-memory observable depend on m consecutive symbols of a finite-state shift. Pass to the higher-block graph whose vertices are admissible $(m - 1)$ -blocks and whose edges are admissible m -blocks. The periodic sum becomes an integer combination of finitely many edge values. A finite collection of such architectures therefore lies in one fixed finite-dimensional rational span V . The edge values need not be algebraic. This certificate does not cover arbitrary Hölder roofs.

5.2 Class M: good reduction gives fixed support

Consider a finite composition of monic area-preserving generalized Hénon factors of degrees at least two over an S -integer ring. Expanding a finite periodic orbit into the individual factors gives a cyclic scalar recurrence

$$z_{j+1} + z_{j-1} = P_{i_j}(z_j). \quad (10)$$

Cyclic neighbor occurrences retain multiplicity: for a one-step cycle, the left side is $2z_0$.

Homogenize the j th equation to its own degree d_{i_j} . On the common hyperplane at infinity, monicity leaves $z_j^{d_{i_j}} = 0$ for every j ; there is no projective point at infinity. After base change to an algebraic closure, the cyclic projective solution scheme therefore lies in an affine chart and is both proper and affine. A finite-type affine scheme proper over a field is finite. Finiteness descends through the field extension, so the cyclic scheme is finite, with zero-dimensional support even when it is nonreduced, and all periodic coordinates are algebraic.

At a non-Archimedean place of good reduction, suppose a cyclic coordinate has maximum norm $R > 1$. At an index attaining the maximum, the monic leading term is uniquely dominant, so $|P_{i_j}(z_j)| = R^{d_{i_j}} > R$. The recurrence gives the contrary bound $|P_{i_j}(z_j)| \leq R$. Hence every periodic coordinate is integral away from the fixed bad places.

The derivative factors and their inverses lie in SL_2 of the integral closure there. Return eigenvalues are therefore units. Put the orbit and eigenvalue in a common normal extension and saturate every place above $S_{\mathbb{Q}}$. Complex conjugation preserves the complement, so for $q = |\lambda| > 0$,

$$q^2 = \lambda \bar{\lambda} \quad (11)$$

is an $S_{\mathbb{Q}}$ -unit. No step identifies $\bar{\lambda}$ with λ^{-1} , and no claim is made that $\log q$ is algebraic.

5.3 Class A: algebraic action with a separate gauge ledger

Let an algebraic exact-symplectic map have a single-valued $\overline{\mathbb{Q}}$ -rational potential G , and let all evaluations on an algebraic periodic orbit be regular. Every value $G(P_j)$ and hence every finite action $\sum_j G(P_j)$ is algebraic. Declared algebraic sums, scales, averages, repetitions, real parts, imaginary parts, and moduli remain real algebraic.

Under a stepwise algebraic exact gauge,

$$G'_j = G_j + \chi_{j+1} \circ F_j - \chi_j + C_j, \quad (12)$$

the action changes by the full endpoint term

$$\chi_n(P_n) - \chi_0(P_0) + \sum_j C_j. \quad (13)$$

Algebraic endpoint shifts preserve algebraicity even when they do not cancel; endpoint compatibility is separately required for canonical gauge invariance. Poles, multivalued logarithmic gauges, closed non-exact primitive changes, transcendental constants, and log-after-action lie outside this source certificate.

5.4 Selector corollary

Corollary 5.1 (Selector L/M/A architecture). *If a realized target is supplied by one L , M , or A component, embed it in (4) as*

$$(v, q, \alpha) = (\ell_L, 1, 0), \quad (0, |\lambda|, 0), \quad (0, 1, \mathcal{A}),$$

respectively. Then (6) applies.

a Source-class certificate ledger

Class L | Locally constant 1/1

proved IDs: L001

fixed finite memory becomes finitely many edge readouts in V

Class M | Multiplier 5/5

proved IDs: M001, M002, M003, M004, M005

$q^2 = \lambda \bar{\lambda}$ is a fixed-support S -unit

Class A | Algebraic action 2/2

proved IDs: A001, A002

regular algebraic action and endpoint shifts stay algebraic

b Terminal upstream closure

	Frozen	Manifest	Pipeline	Record
Paper 3: multiplier	✓	✓	✓	✓
Paper 4: action	✓	✓	✓	✓

c Registered gates (9/9)

Escape semantics	✓	PASS	Exact controls	✓	PASS
Source lock	✓	PASS	Executable isolation	✓	PASS
Independent review	✓	PASS	Upstream bindings	✓	PASS
Proof ledger	✓	PASS	Output scope	✓	PASS
Scope ledger	✓	PASS			

Proof: 20 IDs / 0 cycles

Scope: 10 in / 9 out

Controls: 6 pass

Isolation: 12 files / 0 findings

registered classification: CAPACITY BOUND CERTIFIED | target matches: 0

Figure 3: The three source certificates and their audit boundary. L contributes a vector in one finite rational space; M contributes a real logarithm whose squared modulus has fixed support; A contributes a real algebraic term. The actual terminal Paper-3 and Paper-4 packages validate the upstream provenance, while the registered audit checks the structured dependencies and zero-execution contract.

6 Capacity boundary and positive outside architectures

Corollary 6.1 (Necessary certificate escapes). *An exact construction realizing more than $\dim_{\mathbb{Q}} V + |S_{\mathbb{Q}}|$ distinct rational primes cannot retain all hypotheses of [definition 3.2](#). It must lose the common finite-rank space, fixed finite support, real-algebraic action term, or rational additive normal form.*

This is contraposition within one declared certificate. The four failures may overlap, need not cover arbitrary dynamics, and do not establish an arithmetic correspondence. The identity map on $\mathbb{A}^2$ with constant potential $G \equiv \log 2$ is a useful symbolic control: it shows that a transcendental normalization can inject a target, but it lies outside Class A and evaluates no numerical logarithm. Similarly, allowing $q = p$ with prime-dependent support or inserting $\log p$ as an edge length abandons the target-independence hypothesis rather than refuting the theorem.

Positive arithmetic architectures make the boundary concrete. Deninger’s systems for arithmetic schemes are infinite-dimensional and attach periodic packets to closed points ([Deninger, 2026](#)). Connes–Consani’s scaling site and adèle class space carry prime places intrinsically and exhibit periodic orbits C_p of length $\log p$ ([Connes and Consani, 2016, 2024](#)). These constructions do not fit a frozen finite L/M/A certificate. They are outside examples, not confirmation that one of our four exits is sufficient and not evidence for a universal dimension claim.

The theorem is exact-only. Hermite–Lindemann and valuations do not turn small numerical error into equality, so no approximate separation bound is claimed. Algebraic irrational coefficients on

Table 2: Registered audit outcome. Every value is categorical or exact.

Check	Count/value	Status
Registered gates	9/9	PASS
Proof dependency IDs	20/20	PASS
Admitted / excluded operations	10 / 9	PASS
Exact controls	6/6	PASS
Scanned executable files / findings	12 / 0	PASS
Actual terminal upstream packages	2/2	PASS
Official tests	51/51	PASS
Registered numerical runs / target matches	0 / 0	PASS
Prime tables / target arrays / zero data	false / false / false	PASS

multiplier logs also remain open: the elementary rational-root closure in [lemma 3.3](#) does not cover them.

7 Registered exact/static audit

The implementation was source-locked before the official run. A fresh independent code reviewer bound the exact tree and source-lock hashes with a unique ‘DEPLOYMENT_PASS’ authority. The upstream gate then rehashed seven terminal artifacts for each of the preceding Hénon and algebraic-action papers, including their source locks, proofs, final result manifests, pipeline states, final-integrity records, approved PDFs, and independent Round-2 reviews.

The single registered command performs an exact symbolic/static audit only. All nine gates passed: source lock, independent review, 20-ID proof ledger, scope ledger with ten admitted and nine excluded operations, six exact controls, twelve-file executable isolation, both upstream bindings, escape semantics, and output scope. The post-run suite reported 51 passing tests. The registry records one registered run, zero candidate numerical runs, and zero target matches. The result manifest enforces exact schemas and types, canonical gate recomputation, cross-artifact timestamp and digest agreement, and a flat allowlist.

The audit does not mechanically prove Hermite–Lindemann, the projective-affine dimension lemma, or valuation theory. It verifies that the software uses the reviewed structured dependencies, applies the exact scope, and cannot emit ‘CAPACITY_BOUND_CERTIFIED’ unless every gate is green. The mathematical claims rest on the proofs above and in [appendix B](#).

8 Conclusion and limitations

A finite additive L/M/A certificate has finite arithmetic capacity. Once the locally constant contribution occupies one rational space, multiplier moduli have one finite bad support, and the action contribution is real algebraic, Hermite–Lindemann and finite-place valuations force a rank-plus-support bound on exact rational-prime logarithms. The additive coupling, rather than a selector union, is the central result.

The theorem has deliberate limits. It says nothing quantitative about near matches, does not cover nonlinear or algebraic-irrational multiplier-log mixing, and does not classify nonalgebraic, multivalued, adelic, or infinite-place observables. Its escape conditions are not a complete trichotomy and carry no sufficiency claim. Two concrete next questions are a quantitative analogue for approximate equality and a separately source-locked analysis of an intrinsic outside-class clock. Neither direction is opened by the present static audit.

A Edge-case and scope ledger

Selection and repetitions. For each distinct prime, choose one valid certificate before examining any finite relation. The proof works for every such choice. Multiple orbits or representations for one prime contribute one set element.

Positive rational powers. The unique positive real D th root of a positive algebraic number is algebraic. After adjoining finitely many roots, extension invariance retains the $S_{\mathbb{Q}}$ -unit condition. Negative exponents use inversion. No complex logarithm branch occurs.

Excluded outputs. The certificate forbids output labels asserting a universal symplectic no-go, complete escape trichotomy, infinite-dimensional necessity, historical priority, Riemann-zero progress, or Route-B progress. The accepted classification means only that the locked implication and provenance gates pass.

B Expanded source-class proofs

B.1 The projective-affine step in Class M

For a fixed itinerary, homogenize each equation in (10) to its own degree. At the common homogenizing coordinate $Z = 0$, the monic term forces every cyclic coordinate to vanish; the projective solution scheme has no geometric point at infinity. After base change to an algebraic closure, it is consequently a closed subscheme of an affine chart while remaining proper. A finite-type affine scheme proper over a field is finite, equivalently its coordinate algebra is finite-dimensional over that field. This statement permits nilpotents and does not assume the scheme reduced. Finiteness descends along the field extension. Thus the cyclic scheme is finite. This establishes algebraicity at every finite period, not merely at audited sample periods.

B.2 The non-Archimedean maximum and monodromy

At a good place, all nonleading polynomial coefficients are integral and the leading coefficient is a unit. If $R > 1$ is the maximal cyclic norm, the ultrametric inequality and monicity give a unique leading magnitude $R^d > R$, contradicting the two-neighbor recurrence bound. Hence the orbit is integral. Derivative factors and their inverses have integral entries and determinant one, so a monodromy eigenvalue and its inverse are integral over the local ring. Both valuations vanish. In a normal extension saturated above $S_{\mathbb{Q}}$, the same holds for the conjugate, which proves (11) without a reciprocal-conjugate shortcut.

B.3 Class-A gauge distinction

Summing the stepwise identity gives (13) by ordinary telescoping. If every displayed term is algebraic, the shifted action is algebraic even when $\chi_n(P_n) \neq \chi_0(P_0)$. Endpoint compatibility removes that term and is required only when the numerical closed action is claimed to be gauge invariant. A pole cannot be canceled after evaluation, and monodromy of a multivalued gauge requires a different ledger.

C Reproducibility record

The official sequence was

```
PYTHONPATH=code python code/scripts/run_registered_audit.py
PYTHONPATH=code pytest -q -p no:cacheprovider \
  --junitxml=experiments/official_pytest.xml
PYTHONPATH=code python code/scripts/build_result_manifest.py
```

The corresponding SHA-256 prefixes are

result	9f9878247dc821d1
registry	4ebec117a2254dc45
exclusive manifest	21d6910ec1e8e299
source lock	2d27abceb65cd0ad
reviewed tree	10fd57b1f9961679

The accompanying integrity package records and rechecks every full 64-digit digest.

The figure loader rechecks the source, reviewed tree, official classification, all nine gates, both upstream records, and every zero-target flag. Figure scripts produce PDF and SVG vector masters and PNG review copies with fixed metadata. The registered executable used no network, GPU, floating tolerance, numerical logarithm, prime table, target array, or Riemann-zero data.

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